

Unit 4, Lesson 8: Exploring the Area of a Circle

Benchmark

MA.7.GR.1.4 Explore and apply a formula to find the area of a circle to solve mathematical and real-world problems.

Learning Target

- ☐ I can use the area of a rectangle to develop the formula for the area of a circle.

4.8.1 Warm-Up: Would You Rather?

Table A has a length of 96 inches and a width of 42 inches. Table B has a radius of 45 inches.

1. Which table would you rather use for a large gathering? Explain your choice using mathematics or a picture.

4.8.2 Guided Practice: Area of a Circle

A closed figure is an enclosed shape that occupies space. A circle is an example of a closed figure.

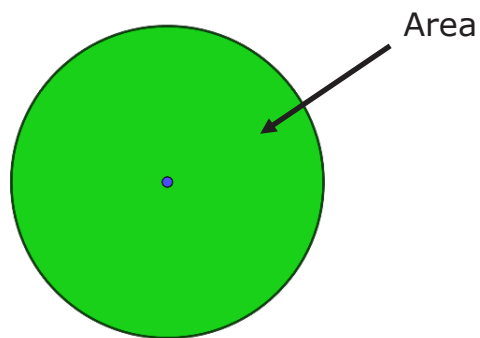
1. Draw a picture of a figure that is not closed.



Area is the measure, in square units, of the inside region of a closed two-dimensional figure.

2. Discuss with your partner why a closed figure can have an area, but a figure that is open cannot have an area. Summarize your discussion.

The measure of the space inside the circle is the circle's area.



4.8.3 Activity: The Formula for the Area of a Circle

1. Your teacher will give you and your partners each a circle that is separated into four, eight, or sixteen sections.
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- Draw an image of your circle and your partners' circles in the first column of the table.
 - Cut out each of the sections of your circle.
 - Rearrange your sections, side by side, to compose a figure that resembles a parallelogram. Paste this in the last column.
 - Draw an image of both of your partners' composed shapes in the last column.

	Circle	Parallelogram
Circle with 4 Sections		
Circle with 8 Sections		
Circle with 16 Sections		

2. Complete the statements.

The circle with 4 sections resembles a

- ☐ square.
- ☐ parallelogram.

The circle with 8 sections resembles a(n)

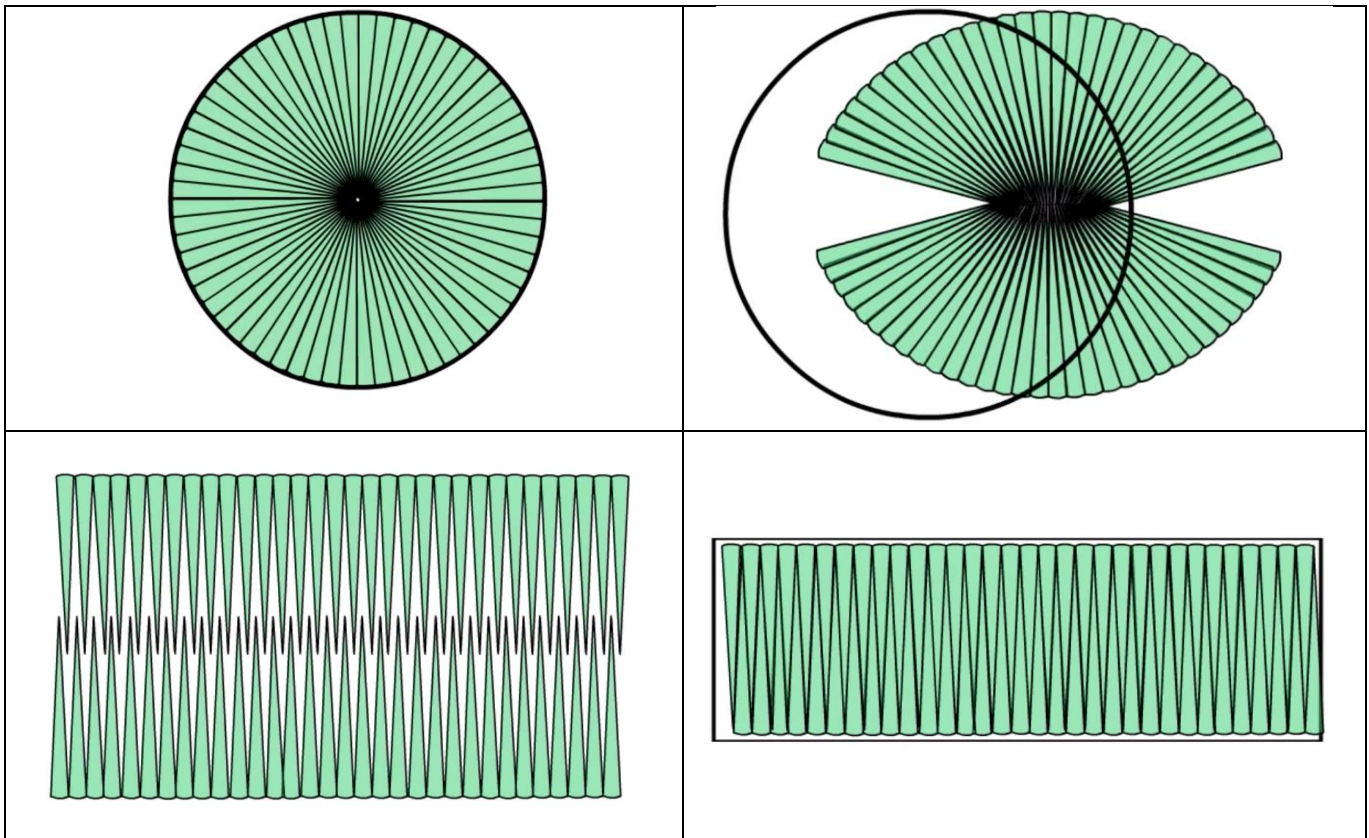
- ☐ square.
- ☐ parallelogram.

The circle with 16 sections most closely resembles a

- ☐ kite.
- ☐ square.
- ☐ rectangle.

3. Discuss with your partners what shape you will see if the number of sectors continues to increase. Summarize your discussion.

This circle has been divided into 64 equal sectors. The sectors were then separated in half and fit together to compose a figure as shown.



4. How many sectors make up the base of the composed shape in the last figure?
5. Trace the base of the composed shape and its corresponding part of the original circle.
6. Trace the height of the shape composed with the 64 sections and the corresponding part of the circle.
7. Complete the following statements.

The formula for the area of a rectangle is

- ☐ $A = b \times h.$
- ☐ $A = 2\pi r.$

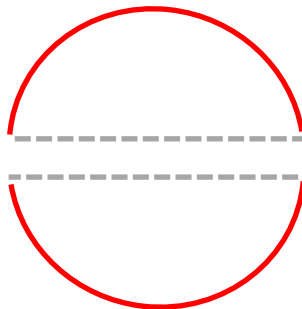
The dimension of the circle that corresponds to one height of the rectangle is

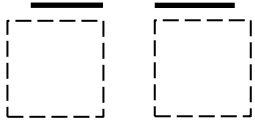
- ☐ the radius.
- ☐ half the circumference.

The dimension of the circle that corresponds to one base of the rectangle is

- ☐ the radius.
- ☐ half the circumference.

8. Given the formula $C = 2\pi r$, work with your partner to find the measure of half the distance around a circle.



$C = 2\pi r$	Formula for the circumference of a circle.
$C = 2\pi r$ 	Divide both sides of the equation by 2.
$\frac{1}{2}C = \underline{\hspace{2cm}}$	Equation written in simplified form for half of the circumference.

9. Given the formula for the area of a rectangle, substitute the corresponding parts of the circle to discover the area of a circle.

$A = b \times h$	Formula for the area of a rectangle.
$A = \underline{\hspace{2cm}} \times \underline{\hspace{2cm}}$	Substitute the parts of the circle that correspond with the base and height of the rectangle.
$A =$	Multiply like bases to write the simplified formula of a circle.

4.8.4 Bring It Together: Area of a Circle

1. Use the words or phrases from the box to complete the sentences.

height rectangle half of the circumference
circle radius sector circumference

The formula for the area of a _____
was used to discover the area of a _____.
When a circle is divided into sectors, the sectors can be
arranged into a rectangle. The base of the rectangle is
_____ of the circle. The height of
the rectangle is the _____ of the
circle. Using this method results in the formula, $A = \frac{1}{2}Cr$.

4.8.5 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate your level of interest in today's learning target and how much it made sense to you.

I can use the area of the rectangle to develop the formula for the area of a circle.

